

A Markov Analysis of Country-Level Social Mobility, 1860-1916

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ChatGPT

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1 Introduction

What follows is a Markov analysis of country-level social mobility with inputs *upper class*, *middle class*, and *lower class* derived from frequency distributions. Histograms were plotted using national primary energy consumption weighted by population with a density overlay. Raw data were taken from the National Materiel Capabilities data set (version 6) hosted by the Correlates of War Project. Primary energy consumption was used as a proxy for industrial output or level of industrialization. Minima in the density overlay were interpreted as class boundaries. Bandwidth was methodologically determined. This approach produced three classes between 1860 and 1880 and four classes between 1881 and 1916. The fourth class in the second era was the result of non-industrial countries developing industrial capabilities and breaking into those markets. It can be seen as a bifurcation of the lower class and an adaptation of its role as the source of cheap labor in the evolving world economy. As the lower class became increasingly industrialized, primary energy consumption was no longer a distinguishing feature of class, and after 1916, this approach to operationalizing class structure was unreliable.

The Markov analysis assessed social mobility, answering the question: *Do poor countries catch up to rich countries?* The question was examined with a balanced panel of countries composed of five-year snapshots between 1860 and 1916. The findings indicate a persistent class structure with *upper class* being an absorbing state in both eras. Structural change was greater in the second era, which is to be expected in moments of hegemonic transition. This can be seen in transitions from *lower class 1* to *lower class 2* (the non-industrial lower class to the industrializing lower class), and in transitions from *middle class* to *lower class 2*, with *lower class 2* being an absorbing state. Thus, structural change in the second era consisted of both upward and downward movement in the economic hierarchy.

An evaluation of Markovian memory rejected independence in favor of first-order dependence, indicating strong state dependence—a country's position at one time point strongly predicts its position at the next. The second order test failed to reject first order sufficiency, not because higher order dependence was absent, but because the high degree of persistence rendered the test uninformative. Over the long interval, Pearson and likelihood-ratio chi-square tests strongly rejected independence, implying substantial long-run persistence in countries' hierarchical positions. The Markov chain failed the test of approximate ergodicity. That is to say, the long-term distribution of countries in the upper, middle, and lower classes was not random. A country's starting position had a lasting effect for the period of examination.

These findings support the contention that economic development, while reducing poverty and improving living conditions, does not change the relative differences of countries in the global economic hierarchy. In this sense, findings undermine the central claim of economic convergence, namely that poor countries are catching up to rich countries.

```
# Preferences
options(scipen = 999,
       digits = 2)
# Package Dependencies
library(dplyr)
library(tidyr)
library(ggplot2)
library(MASS)
library(parallel)
library(readxl)
library(knitr)
library(tibble)
library(stringr)
library(kableExtra)

# Data Import and Recoding
```

```
df <- read_excel("Class Assignments.xlsx") %>%
  dplyr::select(stateabb, year, class) %>%
  dplyr::mutate(class = dplyr::recode(class,
                                     "core" = "upper_class",
                                     "semiperiphery" = "middle_class",
                                     "periphery_1" = "lower_class_1",
                                     "periphery_2" = "lower_class_2"))
```

2 Transition Sequence Construction

```
# Function for computing transitions
compute_trans <- function(data,
                           snapshot_years,
                           step,
                           full_levels) {

  df_snap <- data %>%
    dplyr::filter(year %in% snapshot_years) %>%
    dplyr::mutate(class = factor(class, levels = full_levels))
  df_balanced <- df_snap %>%
    arrange(stateabb, year) %>%
    group_by(stateabb) %>%
    mutate(n_obs = n()) %>%
    ungroup()
  message("Balanced-panel countries for snapshots {",
          paste(snapshot_years, collapse = ", "),
          "}: ", paste("n =",
                       length(unique(df_balanced$stateabb))))
  df_trans <- df_balanced %>%
    dplyr::arrange(stateabb, year) %>%
    dplyr::group_by(stateabb) %>%
    dplyr::mutate(
      class_prev = dplyr::lag(class),
      year_prev = dplyr::lag(year)) %>%
    dplyr::ungroup() %>%
    dplyr::filter(
      !is.na(class_prev),
      (year - year_prev) %in% step) %>%
    dplyr::mutate(
      step_id = factor(paste0(year_prev, "->", year)))
  df_trans
}

# Era 1: 1860-1880
# 3-class system: upper_class, middle_class, lower_class_1
# Snapshots: 1860, 1865, 1870, 1875, 1880 (5-year steps)

# Subset data
era1_years <- 1860:1880
df1 <- df %>% dplyr::filter(year %in% era1_years)

levels_era1 <- c("upper_class", "middle_class", "lower_class_1")
```

```

snap1 <- c(1860, 1865, 1870, 1875, 1880)
step1 <- 5

# Compute & save 5-year transitions
trans1 <- compute_trans(
  data      = df1,
  snapshot_years = snap1,
  step      = step1,
  full_levels = levels_era1)

# Era 2: (1881-1916)
# 3-class system: upper_class, middle_class, lower_class_2, lower_class_1
# Snapshots: 1881, 1886, 1891, 1896, 1901, 1906, 1911, 1916 (5-year steps)

# Subset data
era2_years <- 1881:1916
df2 <- df %>% dplyr::filter(year %in% era2_years)

levels_era2 <- c("upper_class", "middle_class", "lower_class_2", "lower_class_1")

snap2 <- c(1881, 1886, 1891, 1896, 1901, 1906, 1911, 1916)
step2 <- 5

# Compute & save 5-year transitions
trans2 <- compute_trans(
  data      = df2,
  snapshot_years = snap2,
  step      = step2,
  full_levels = levels_era2)

# Data set Overview
era_summary <- tibble::tibble(
  Era = c("Era 1 (1860-1880)", "Era 2 (1881-1916)"),
  Snapshots = c(
    paste(snap1, collapse = ", "),
    paste(snap2, collapse = ", ")),
  Transitions = c(
    nrow(trans1),
    nrow(trans2)))

# Table header
cat('Balanced-panel snapshot coverage and transition counts')
era_summary_ordered <- era_summary %>%
  dplyr::select(Era, Transitions, Snapshots)

# Balanced panel summary with kable
knitr::kable(
  era_summary_ordered,
  format      = "latex",
  booktabs   = TRUE,
  caption     = "Summary of Markov Order Tests by Era") %>%
  kableExtra::kable_styling(
    latex_options = "hold_position"

```

)

Balanced-panel snapshot coverage and transition counts

Table 1: Summary of Markov Order Tests by Era

Era	Transitions	Snapshots
Era 1 (1860-1880)	138	1860, 1865, 1870, 1875, 1880
Era 2 (1881-1916)	274	1881, 1886, 1891, 1896, 1901, 1906, 1911, 1916

3 Sample Attrition Diagnostic

```
# Systematic attrition test functions
attrition_diagnostic <- function(data,
                                snapshot_years,
                                balanced_df,
                                full_levels) {
  kept_states <- unique(balanced_df$stateabb)
  year0 <- min(snapshot_years)
  all_year0 <- data %>%
    dplyr::filter(year == year0) %>%
    dplyr::mutate(class = factor(class, levels = full_levels)) %>%
    dplyr::count(class, name = "n") %>%
    dplyr::mutate(group = "All")
  kept_year0 <- data %>%
    dplyr::filter(year == year0, stateabb %in% kept_states) %>%
    dplyr::mutate(class = factor(class, levels = full_levels)) %>%
    dplyr::count(class, name = "n") %>%
    dplyr::mutate(group = "Balanced")
  df_comp <- dplyr::bind_rows(all_year0, kept_year0)
  tab <- xtabs(n ~ group + class, data = df_comp)
  chisq <- suppressWarnings(chisq.test(tab))
  list(counts = df_comp,
       table = tab,
       chisq = chisq)
}

# Print function for  $\chi^2$ 
print_chisq_compact <- function(chisq_obj) {
  cat("X^2 =", round(chisq_obj$statistic, 3),
      " df =", chisq_obj$parameter,
      " p =", signif(chisq_obj$p.value, 4), "\n")
}

# Interpretation guide
interpret_attrition <- function(chisq_obj, alpha = 0.05) {
  p <- chisq_obj$p.value
  if (p < alpha) {
    paste0("Reject equality of baseline class distributions (p < ", alpha,
          "). Evidence of selective attrition.")
  } else {
```

```

    paste0("Fail to reject equality of baseline class distributions (p ", alpha,
           "). No evidence of selective attrition.")
  }
}

# Filter at five-year intervals
df1_snap <- df1 %>% dplyr::filter(year %in% snap1)
df2_snap <- df2 %>% dplyr::filter(year %in% snap2)

# Build era 1 sample
df1_balanced <- df1_snap %>%
  dplyr::group_by(stateabb) %>%
  dplyr::filter(dplyr::n() == length(snap1)) %>%
  dplyr::ungroup()

df2_balanced <- df2_snap %>%
  dplyr::group_by(stateabb) %>%
  dplyr::filter(dplyr::n() == length(snap2)) %>%
  dplyr::ungroup()

# Test for systematic attrition
attr1 <- attrition_diagnostic(df1, snap1, df1_balanced, levels_era1)
attr2 <- attrition_diagnostic(df2, snap2, df2_balanced, levels_era2)

# Function for PDF output
render_attrition_block <- function(attr, era_label, era_years) {
  print(
    knitr::kable(
      attr$counts,
      format = "latex",
      booktabs = TRUE,
      caption = paste0(
        era_label, " (", era_years,
        "): Observed Counts"
      ) %>%
      kableExtra::kable_styling(latex_options = "hold_position")
    )
  )
  cat("\n")
  cat("\textbf{Test Statistic}\n\n")
  chi <- attr$chisq
  cat(sprintf(
    "\textit{Chi-square test}: $\chi^2(%d)=%.3f$, $p=.%4f$.\n\n",
    unname(chi$parameter),
    unname(chi$statistic),
    unname(chi$p.value))
  )
  cat("\textbf{Interpretation}\n\n")
  cat(interpret_attrition(attr$chisq), "\n\n")
  cat("\bigskip\n")
  invisible(NULL)
}

```

3.1 Era 1 1860-1880

```
render_attrition_block(attr1, "Era 1", "1860--1880")
```

Table 2: Era 1 (1860–1880): Observed Counts

class	n	group
upper_class	8	All
middle_class	16	All
lower_class_1	13	All
upper_class	6	Balanced
middle_class	14	Balanced
lower_class_1	12	Balanced

Test Statistic

Chi-square test: $\chi^2(2) = 0.097$, $p = 0.9525$.

Interpretation

Fail to reject equality of baseline class distributions ($p = 0.05$). No evidence of selective attrition.

3.2 Era 2 1881-1916

```
render_attrition_block(attr2, "Era 2", "1881--1916")
```

Table 3: Era 2 (1881–1916): Observed Counts

class	n	group
upper_class	7	All
middle_class	16	All
lower_class_2	1	All
lower_class_1	13	All
upper_class	7	Balanced
middle_class	16	Balanced
lower_class_2	1	Balanced
lower_class_1	9	Balanced

Test Statistic

Chi-square test: $\chi^2(3) = 0.500$, $p = 0.9188$.

Interpretation

Fail to reject equality of baseline class distributions ($p = 0.05$). No evidence of selective attrition.

4 5-Year Step Transition Matrices

```
# Build transition counts (Era 1)
C1 <- xtabs(~ class_prev + class, data = trans1)
C1 <- C1[levels_era1, levels_era1, drop = FALSE]
C1_df <- as.data.frame.matrix(C1)
```

```

# Build transition probabilities (Era 1)
P1 <- prop.table(C1, margin = 1)
P1_df <- as.data.frame.matrix(round(P1, 3))

# Build transition counts (Era 2)
C2 <- xtabs(~ class_prev + class, data = trans2)
C2 <- C2[levels_era2, levels_era2, drop = FALSE]
C2_df <- as.data.frame.matrix(C2)

# Build transition probabilities (Era 2)
P2 <- prop.table(C2, margin = 1)
P2_df <- as.data.frame.matrix(round(P2, 3))

```

4.1 Era 1 1860-1880

```

## Era 1 (1860-1880)

# Count transition matrix with kable
knitr::kable(
  C1_df,
  format = "latex",
  booktabs = TRUE,
  caption = "Era 1 (1860-1880): First-Order Transition Counts (5-Year Steps)" %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")

# Probability transition matrix with kable
knitr::kable(
  round(P1_df, 3),
  format = "latex",
  booktabs = TRUE,
  caption = "Era 1 (1860-1880): First-Order Transition Probabilities (5-Year Steps)" %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")

```

Table 4: Era 1 (1860–1880): First-Order Transition Counts (5-Year Steps)

	upper_class	middle_class	lower_class_1
upper_class	28	1	0
middle_class	1	57	0
lower_class_1	0	1	50

Table 5: Era 1 (1860–1880): First-Order Transition Probabilities (5-Year Steps)

	upper_class	middle_class	lower_class_1
upper_class	0.97	0.03	0.00
middle_class	0.02	0.98	0.00
lower_class_1	0.00	0.02	0.98

4.2 Era 2 1881-1916

```
## Era 2 (1881-1916)

# Count transition matrix with kable
knitr::kable(
  C2_df,
  format = "latex",
  booktabs = TRUE,
  caption = "Era 2 (1881-1916): First-Order Transition Counts (5-Year Steps)" %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")

# Probability transition matrix with kable
knitr::kable(
  round(P2_df, 3),
  format = "latex",
  booktabs = TRUE,
  caption = "Era 2 (1881-1916): First-Order Transition Probabilities (5-Year Steps)" %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")
```

Table 6: Era 2 (1881–1916): First-Order Transition Counts (5-Year Steps)

	upper_class	middle_class	lower_class_2	lower_class_1
upper_class	36	3	0	0
middle_class	3	100	8	0
lower_class_2	0	3	41	0
lower_class_1	0	0	6	74

Table 7: Era 2 (1881–1916): First-Order Transition Probabilities (5-Year Steps)

	upper_class	middle_class	lower_class_2	lower_class_1
upper_class	0.92	0.08	0.00	0.00
middle_class	0.03	0.90	0.07	0.00
lower_class_2	0.00	0.07	0.93	0.00
lower_class_1	0.00	0.00	0.07	0.92

5 5-Year Step Time Homogeneity Tests

```
# Function to test time homogeneity of 5-year steps
test_5_step_homog <- function(transitions_df) {
  counts <- transitions_df %>%
    dplyr::count(step_id, class_prev, class, name = "n") %>%
    tidyr::complete(step_id, class_prev, class, fill = list(n = 0))
  df_counts <- as.data.frame(counts)
  colnames(df_counts) <- c("step", "origin", "dest", "n")
  fit_homog <- MASS::loglm(
    n ~ origin + dest + step + origin:dest,
```

```

    data = df_counts)
fit_full <- MASS::loglm(
  n ~ origin * dest * step,
  data = df_counts)
G2      <- fit_homog$deviance - fit_full$deviance
df_diff <- fit_homog$df - fit_full$df
p_val   <- 1 - pchisq(G2, df_diff)
list(
  counts    = df_counts,
  G2        = G2,
  df        = df_diff,
  p_value   = p_val,
  fit_homog = fit_homog,
  fit_full  = fit_full)
}

# Interpretation Guide
interpret_homogeneity <- function(p, alpha = 0.05) {
  if (p < alpha) {
    paste0("Reject homogeneity. Evidence of structural change.")
  } else {
    paste0("Fail to reject homogeneity. No evidence of structural change.")
  }
}

short_homog_era_1 <- test_5_step_homog(trans1)
short_homog_era_2 <- test_5_step_homog(trans2)

homog_5_step_results <- tibble(
  Era = c("1860-1880", "1881-1916"),
  `G^2` = c(short_homog_era_1$G2, short_homog_era_2$G2),
  df = c(short_homog_era_1$df, short_homog_era_2$df),
  `p-value` = c(short_homog_era_1$p_value, short_homog_era_2$p_value),
  Interpretation = c(
    interpret_homogeneity(short_homog_era_1$p_value),
    interpret_homogeneity(short_homog_era_2$p_value))) %>%
  mutate(
    `G^2` = round(`G^2`, 3),
    `p-value` = signif(`p-value`, 4))

```

5.1 Results

```

homog_5_step_results %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Short-Interval Markov Time Homogeneity Tests") %>%
  kable_styling(
    latex_options = c("hold_position"),
    font_size = 10)

```

Table 8: Short-Interval Markov Time Homogeneity Tests

Era	G^2	df	p-value	Interpretation
1860–1880	8.5	24	1.00	Fail to reject homogeneity. No evidence of structural change.
1881–1916	63.2	90	0.99	Fail to reject homogeneity. No evidence of structural change.

6 Long-Interval Transition Matrices

```
# Function for computing long-interval transitions
compute_long_trans <- function(data,
                                start_year,
                                end_year,
                                full_levels) {

  df_start <- data %>%
    dplyr::filter(year == start_year) %>%
    dplyr::select(stateabb, class_start = class)
  df_end <- data %>%
    dplyr::filter(year == end_year) %>%
    dplyr::select(stateabb, class_end = class)
  df_long <- dplyr::inner_join(df_start, df_end, by = "stateabb") %>%
    dplyr::mutate(
      class_start = factor(class_start, levels = full_levels),
      class_end = factor(class_end, levels = full_levels)) %>%
    dplyr::filter(!is.na(class_start), !is.na(class_end))
  tab <- with(df_long, table(class_start, class_end))
  list(
    df_long = df_long,
    tab = tab)
}

long_interval_probs <- function(tab) {
  prop.table(tab, margin = 1)
}

# Compute transitions for era 1 (1860 + 1880)
long1 <- compute_long_trans(
  data = df1,
  start_year = 1860,
  end_year = 1880,
  full_levels = levels_era1)

# Save counts table
long1$tab <- long1$tab[levels_era1, levels_era1, drop = FALSE]

# Save probabilities table
P_long1 <- long_interval_probs(long1$tab)

# Compute transitions for era 2 (1881 + 1916)
long2 <- compute_long_trans(
  data = df2,
  start_year = 1881,
  end_year = 1916,
```

```

full_levels = levels_era2)

# Save counts table
long2$tab <- long2$tab[levels_era2, levels_era2, drop = FALSE]

# Save probabilities table
P_long2 <- long_interval_probs(long2$tab)

```

6.1 Era 1 1860-1880

```

# Count transition matrix with kable
knitr::kable(
  long1$tab,
  format = "latex",
  booktabs = TRUE,
  caption = "Era 1 Long-Interval Transition Counts (1860 → 1880)" %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")

# Probabilitt transition matrix with kable
knitr::kable(
  round(P_long1, 3),
  format = "latex",
  booktabs = TRUE,
  caption = "Era 1 Long-Interval Transition Probabilities (1860 → 1880)" %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")

```

Table 9: Era 1 Long-Interval Transition Counts (1860 → 1880)

	upper_class	middle_class	lower_class_1
upper_class	6	0	0
middle_class	0	14	0
lower_class_1	0	1	12

Table 10: Era 1 Long-Interval Transition Probabilities (1860 → 1880)

	upper_class	middle_class	lower_class_1
upper_class	1	0.00	0.00
middle_class	0	1.00	0.00
lower_class_1	0	0.08	0.92

6.2 Era 2 1881-1916

```

# Counts transition matrix with kable
knitr::kable(
  long2$tab,
  format = "latex",
  booktabs = TRUE,

```

```

caption = "Era 2 Long-Interval Transition Counts (1881 → 1916)" %>%
kableExtra::kable_styling(
  latex_options = "hold_position")

# Probabilities transition matrix with kable
knitr::kable(
  round(P_long2, 3),
  format = "latex",
  booktabs = TRUE,
  caption = "Era 2 Long-Interval Transition Probabilities (1881 → 1916)" %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")

```

Table 11: Era 2 Long-Interval Transition Counts (1881 → 1916)

	upper_class	middle_class	lower_class_2	lower_class_1
upper_class	7	0	0	0
middle_class	0	11	5	0
lower_class_2	0	0	1	0
lower_class_1	0	0	5	4

Table 12: Era 2 Long-Interval Transition Probabilities (1881 → 1916)

	upper_class	middle_class	lower_class_2	lower_class_1
upper_class	1	0.00	0.00	0.00
middle_class	0	0.69	0.31	0.00
lower_class_2	0	0.00	1.00	0.00
lower_class_1	0	0.00	0.56	0.44

7 Long-Interval Transition Diagnostics

```

# Bowker symmetry test
bowker_symmetry_safe <- function(tab) {
  tab <- as.matrix(tab)
  if (nrow(tab) != ncol(tab)) stop("tab must be square")
  k <- nrow(tab)
  X2 <- 0
  df <- 0
  for (i in 1:(k - 1)) {
    for (j in (i + 1):k) {
      nij <- tab[i, j]; nji <- tab[j, i]
      s <- nij + nji
      if (s > 0) {
        X2 <- X2 + (nij - nji)^2 / s
        df <- df + 1
      }
    }
  }
  p <- if (df > 0) 1 - pchisq(X2, df) else NA_real_
}

```

```

    list(X2 = X2, df = df, p = p)
  }

# Bowker  $X^2$ , Pearson  $X^2$  & Likelihood Ratio  $X^2$  long-interval tests
db_long_interval_tests <- function(tab) {
  pear <- suppressWarnings(stats::chisq.test(tab, correct = FALSE))
  df_counts <- as.data.frame(tab)
  colnames(df_counts) <- c("origin", "dest", "n")
  fit_ind <- MASS::loglm(n ~ origin + dest, data = df_counts)
  LR_G2 <- fit_ind$deviance
  LR_df <- fit_ind$df
  LR_p <- 1 - pchisq(LR_G2, LR_df)
  bow <- bowker_symmetry_safe(tab)
  list(
    Pearson = list(stat = unname(pear$statistic), df = unname(pear$parameter), p = pear$p.value),
    LR      = list(stat = LR_G2, df = LR_df, p = LR_p),
    Bowker  = list(stat = bow$X2, df = bow$df, p = bow$p))
}

# Interpretation Guide
interp_indep <- function(p, alpha = 0.05) {
  if (is.na(p)) return("Not estimable.")
  if (p < alpha) "Reject independence. Strong persistence." else "Fail to reject independence."
}

interp_sym <- function(p, alpha = 0.05) {
  if (is.na(p)) return("Not estimable.")
  if (p < alpha) "Reject symmetry. Directional movement." else "Fail to reject symmetry."
}

p_fmt <- function(p) {
  if (is.na(p)) return("NA")
  if (p < 1e-4) return("<0.01")
  formatC(p, format = "f", digits = 2)
}

# Knit with kable
make_long_interval_summary <- function(era_label, tab, alpha = 0.05) {
  t <- db_long_interval_tests(tab)
  dplyr::bind_rows(
    tibble::tibble(
      Era = era_label,
      Test = "Pearson  $X^2$ ",
      Statistic = round(t$Pearson$stat, 2),
      df = t$Pearson$df,
      `p-value` = p_fmt(t$Pearson$p),
      Interpretation = interp_indep(t$Pearson$p, alpha)),
    tibble::tibble(
      Era = era_label,
      Test = "Likelihood Ratio  $X^2$ ",
      Statistic = round(t$LR$stat, 2),
      df = t$LR$df,
      `p-value` = p_fmt(t$LR$p),

```

```

    Interpretation = interp_indep(t$LR$p, alpha)),
  tibble::tibble(
    Era = era_label,
    Test = "Bowker X2",
    Statistic = round(t$Bowker$stat, 2),
    df = t$Bowker$df,
    `p-value` = p_fmt(t$Bowker$p),
    Interpretation = interp_sym(t$Bowker$p, alpha)))
}

```

7.1 Era 1 1860-1880

```

summary_tbl_era1 <- make_long_interval_summary("1860-1880", long1$tab, alpha = 0.05)

knitr::kable(
  summary_tbl_era1,
  format = "latex",
  booktabs = TRUE,
  caption = "ERA 1 (1860-1880): Long-interval diagnostic tests") %>%
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"), font_size = 9) %>%
  kableExtra::row_spec(0, bold = TRUE) %>%
  print()

```

Table 13: ERA 1 (1860–1880): Long-interval diagnostic tests

Era	Test	Statistic	df	p-value	Interpretation
1860–1880	Pearson X ²	61	4	<0.01	Reject independence. Strong persistence.
1860–1880	Likelihood Ratio X ²	61	4	<0.01	Reject independence. Strong persistence.
1860–1880	Bowker X ²	1	1	0.32	Fail to reject symmetry.

7.2 Era 2 1881-1916

```

summary_tbl_era2 <- make_long_interval_summary("1881-1916", long2$tab, alpha = 0.05)

knitr::kable(
  summary_tbl_era2,
  format = "latex",
  booktabs = TRUE,
  caption = "ERA 2 (1881-1916): Long-interval diagnostic tests") %>%
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"), font_size = 9) %>%
  kableExtra::row_spec(0, bold = TRUE) %>%
  print()

```

Table 14: ERA 2 (1881–1916): Long-interval diagnostic tests

Era	Test	Statistic	df	p-value	Interpretation
1881–1916	Pearson X ²	53	9	<0.01	Reject independence. Strong persistence.
1881–1916	Likelihood Ratio X ²	55	9	<0.01	Reject independence. Strong persistence.
1881–1916	Bowker X ²	10	2	0.01	Reject symmetry. Directional movement.

8 Higher-Order Dependency Structure

```
# Function for building Markov pairs
build_markov_pairs <- function(data,
                               snapshot_years,
                               step,
                               full_levels) {
  df_snap <- data %>%
    dplyr::filter(year %in% snapshot_years) %>%
    dplyr::mutate(class = factor(class, levels = full_levels))
  df_pairs <- df_snap %>%
    dplyr::arrange(stateabb, year) %>%
    dplyr::group_by(stateabb) %>%
    dplyr::mutate(class_lag1 = dplyr::lag(class, 1),
                  year_lag1 = dplyr::lag(year, 1)) %>%
    dplyr::ungroup() %>%
    dplyr::filter(!is.na(class_lag1),
                  (year - year_lag1) %in% step) %>%
    dplyr::rename(class_curr = class)
  df_pairs
}

# Function for building Markov triples
build_markov_triples <- function(data,
                                  snapshot_years,
                                  step,
                                  full_levels) {
  df_snap <- data %>%
    dplyr::filter(year %in% snapshot_years) %>%
    dplyr::mutate(class = factor(class, levels = full_levels))
  df_with_counts <- df_snap %>%
    dplyr::arrange(stateabb, year) %>%
    dplyr::group_by(stateabb) %>%
    dplyr::mutate(n_obs = dplyr::n()) %>%
    dplyr::ungroup()
  df_triple <- df_with_counts %>%
    dplyr::arrange(stateabb, year) %>%
    dplyr::group_by(stateabb) %>%
    dplyr::mutate(class_lag1 = dplyr::lag(class, 1),
                  year_lag1 = dplyr::lag(year, 1),
                  class_lag2 = dplyr::lag(class, 2),
                  year_lag2 = dplyr::lag(year, 2)) %>%
    dplyr::ungroup() %>%
    dplyr::filter(!is.na(class_lag1),
                  !is.na(class_lag2),
                  (year - year_lag1) %in% step,
                  (year_lag1 - year_lag2) %in% step) %>%
    dplyr::rename(class_curr = class)
}

# Era 1: 1860-1880 Build Markov pairs & triples
pairs1 <- build_markov_pairs(
  data      = df1,
  snapshot_years = snap1,
```

```

step          = step1,
full_levels   = levels_era1)

triples1 <- build_markov_triples(
  data          = df1,
  snapshot_years = snap1,
  step          = step1,
  full_levels   = levels_era1)

# Era 2: 1881-1916 Build Markov pairs & triples
pairs2 <- build_markov_pairs(
  data          = df2,
  snapshot_years = snap2,
  step          = step2,
  full_levels   = levels_era2)

triples2 <- build_markov_triples(
  data          = df2,
  snapshot_years = snap2,
  step          = step2,
  full_levels   = levels_era2)

```

9 Evaluation of Markovian Memory

```

# Testing function for first order pairs
test_first_order <- function(pairs_df, full_levels) {
  df_counts <- pairs_df %>%
    dplyr::count(class_lag1, class_curr, name = "n") %>%
    dplyr::mutate(class_lag1 = factor(class_lag1, levels = full_levels),
                  class_curr = factor(class_curr, levels = full_levels)) %>%
    tidyr::complete(class_lag1, class_curr, fill = list(n = 0))
  colnames(df_counts) <- c("M", "N", "n")
  fit_H0 <- MASS::loglm(n ~ M + N, data = df_counts) # 0th order (indep)
  fit_H1 <- MASS::loglm(n ~ M * N, data = df_counts) # 1st order
  G2 <- fit_H0$deviance - fit_H1$deviance
  df_diff <- fit_H0$df - fit_H1$df
  p_val <- 1 - pchisq(G2, df_diff)
  list(counts = df_counts,
       G2 = G2,
       df = df_diff,
       p_value = p_val,
       fit_H0 = fit_H0,
       fit_H1 = fit_H1)
}

# Testing function for second order triples
test_second_order <- function(triples_df, full_levels) {
  df_counts <- triples_df %>%
    dplyr::count(class_lag2, class_lag1, class_curr, name = "n") %>%
    dplyr::mutate(class_lag2 = factor(class_lag2, levels = full_levels),
                  class_lag1 = factor(class_lag1, levels = full_levels),
                  class_curr = factor(class_curr, levels = full_levels)) %>%

```

```

  tidyr::complete(class_lag2, class_lag1, class_curr, fill = list(n = 0))
  colnames(df_counts) <- c("L", "M", "N", "n")
  fit_H0 <- MASS::loglm(n ~ L + M + N + L:M + M:N, data = df_counts) # 1st-order Markov
  fit_H1 <- MASS::loglm(n ~ L * M * N, data = df_counts) # 2nd-order (saturated)
  G2 <- fit_H0$deviance - fit_H1$deviance
  df_diff <- fit_H0$df - fit_H1$df
  p_val <- 1 - pchisq(G2, df_diff)
  list(counts = df_counts,
       G2 = G2,
       df = df_diff,
       p_value = p_val,
       fit_H0 = fit_H0,
       fit_H1 = fit_H1)
}

# Interpretation Guide
interpret_markov_order <- function(p, comparison, alpha = 0.05) {

  if (comparison == "0vs1") {
    if (p < alpha) {
      return(paste0("Reject independence.
                     Evidence of first-order state dependence."))
    } else {
      return(paste0("Fail to reject zero-order assumption.
                     Transitions consistent with independence."))
    }
  }

  if (comparison == "1vs2") {
    if (p < alpha) {
      return(paste0("Reject first-order sufficiency.
                     Evidence of second-order dependence."))
    } else {
      return(paste0("Fail to reject first-order sufficiency.
                     First-order Markov assumption adequate."))
    }
  }

  stop("Unknown comparison type: use '0vs1' or '1vs2'")
}

# Run and save tests for Era 1 (1860-1880)
test1_markov01 <- test_first_order(pairs1, levels_era1)
test1_markov12 <- test_second_order(triples1, levels_era1)

# Run and save tests for Era 2 (1881-1916)
test2_markov01 <- test_first_order(pairs2, levels_era2)
test2_markov12 <- test_second_order(triples2, levels_era2)

# Preprocessing for kable
markov_order_results <- tibble(
  Era = c(
    "1860-1880",

```

```

"1860-1880",
"1881-1916",
"1881-1916"),
Comparison = c(
  "Zero vs First-order",
  "First vs Second-order",
  "Zero vs First-order",
  "First vs Second-order"),
`G^2` = c(
  test1_markov01$G2, # Era 1, 0 vs 1
  test1_markov12$G2, # Era 1, 1 vs 2
  test2_markov01$G2, # Era 2, 0 vs 1
  test2_markov12$G2, # Era 2, 1 vs 2),
df = c(
  test1_markov01$df,
  test1_markov12$df,
  test2_markov01$df,
  test2_markov12$df),
`p-value` = c(
  test1_markov01$p_value,
  test1_markov12$p_value,
  test2_markov01$p_value,
  test2_markov12$p_value),
Interpretation = c(
  interpret_markov_order(test1_markov01$p_value, "0vs1"),
  interpret_markov_order(test1_markov12$p_value, "1vs2"),
  interpret_markov_order(test2_markov01$p_value, "0vs1"),
  interpret_markov_order(test2_markov12$p_value, "1vs2"))) %>%
mutate(
  `G^2` = round(`G^2`, 3),
  `p-value` = signif(`p-value`, 4),
  Interpretation = stringr::str_wrap(Interpretation, width = 45))

markov_01_results <- markov_order_results %>%
  dplyr::filter(Comparison == "Zero vs First-order") %>%
  dplyr::select(-Comparison)

markov_12_results <- markov_order_results %>%
  dplyr::filter(Comparison == "First vs Second-order") %>%
  dplyr::select(-Comparison)

```

9.1 Results

```

# First order test results with kable
knitr::kable(
  markov_01_results,
  format = "latex",
  booktabs = TRUE,
  caption = "First Order Test: Zero vs First-Order Dependence") %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")

# Second order test results with kable

```

```
knitr::kable(
  markov_12_results,
  format = "latex",
  booktabs = TRUE,
  caption = "Second Order Test: First vs Second-Order Dependence") %>%
  kableExtra::kable_styling(
    latex_options = "hold_position")
```

Table 15: First Order Test: Zero vs First-Order Dependence

Era	G^2	df	p-value	Interpretation
1860–1880	264	4	0	Reject independence. Evidence of first-order state dependence.
1881–1916	553	9	0	Reject independence. Evidence of first-order state dependence.

Table 16: Second Order Test: First vs Second-Order Dependence

Era	G^2	df	p-value	Interpretation
1860–1880	0.1	12	1.00	Fail to reject first-order sufficiency. First-order Markov assumption adequate.
1881–1916	21.1	36	0.98	Fail to reject first-order sufficiency. First-order Markov assumption adequate.

10 Ergodicity Diagnostic

```
# Function for testing ergodicity
check_ergodicity <- function(P, max_power = 10, tol = 1e-12) {
  A <- (P > tol) * 1
  A_power <- A
  reachable_all <- FALSE
  for (k in 1:max_power) {
    if (all(A_power > 0)) {
      reachable_all <- TRUE
      break
    }
    A_power <- (A_power %*% A > 0) * 1
  }
  P_power <- P
  positive_diag_all <- FALSE
  for (k in 1:max_power) {
    if (all(diag(P_power) > tol)) {
      positive_diag_all <- TRUE
      break
    }
    P_power <- P_power %*% P
  }
  list(
    irreducible_approx = reachable_all,
    aperiodic_approx = positive_diag_all,
    max_power_checked = max_power)
}
```

```

# Interpretation guide
interpret_ergodicity <- function(erg) {
  if (isTRUE(erg$irreducible_approx) && isTRUE(erg$aperiodic_approx)) {
    return("Chain is ergodic (approximately irreducible and aperiodic).")
  }
  if (isFALSE(erg$irreducible_approx) && isTRUE(erg$aperiodic_approx)) {
    return("Chain is not ergodic (fails approximate irreducibility).")
  }
  if (isTRUE(erg$irreducible_approx) && isFALSE(erg$aperiodic_approx)) {
    return("Chain is not ergodic (fails approximate aperiodicity).")
  }
  return("Ergodicity status unclear under approximation criteria.")
}

# Function for PDF output
ergodicity_to_df <- function(erg) {
  data.frame(
    criterion = c("Approx. irreducible", "Approx. aperiodic", "Max power checked"),
    value     = c(
      ifelse(isTRUE(erg$irreducible_approx), "Yes", "No"),
      ifelse(isTRUE(erg$aperiodic_approx),   "Yes", "No"),
      erg$max_power_checked),
    stringsAsFactors = FALSE)
}

# Run test and save
erg2 <- check_ergodicity(P2)
erg1 <- check_ergodicity(P1)

```

10.1 Era 1 1860-1880

```

# Table first
print(
  knitr::kable(
    ergodicity_to_df(erg1),
    format      = "latex",
    booktabs    = TRUE,
    caption     = "Era 1 (1860-1880): Ergodicity Diagnostic (Approximation Criteria)" %>%
    kableExtra::kable_styling(latex_options = "hold_position"))

```

Table 17: Era 1 (1860–1880): Ergodicity Diagnostic (Approximation Criteria)

criterion	value
Approx. irreducible	No
Approx. aperiodic	Yes
Max power checked	10

Interpretation: Chain is not ergodic (fails approximate irreducibility).

10.2 Era 2 1881-1916

```
print(
  knitr::kable(
    ergodicity_to_df(erg2),
    format = "latex",
    booktabs = TRUE,
    caption = "Era 2 (1881-1916): Ergodicity Diagnostic (Approximation Criteria)" %>%
    kableExtra::kable_styling(latex_options = "hold_position"))
```

Table 18: Era 2 (1881–1916): Ergodicity Diagnostic (Approximation Criteria)

criterion	value
Approx. irreducible	No
Approx. aperiodic	Yes
Max power checked	10

Interpretation: Chain is not ergodic (fails approximate irreducibility).

11 Comparison of Structural Change Across Eras

```
# Equality of Transition Matrices Between Eras
compare_eras_markov <- function(trans1_df, trans2_df,
                                era1_label = "Era1",
                                era2_label = "Era2",
                                common_levels = c("upper_class", "middle_class", "lower_class_1")) {

  prep <- function(df, era) {
    df %>%
      dplyr::mutate(
        class_prev = factor(class_prev, levels = common_levels),
        class = factor(class, levels = common_levels)) %>%
      dplyr::filter(!is.na(class_prev), !is.na(class)) %>%
      dplyr::count(class_prev, class, name = "n") %>%
      dplyr::mutate(era = era)
  }

  df_counts <- dplyr::bind_rows(
    prep(trans1_df, era1_label),
    prep(trans2_df, era2_label))
  colnames(df_counts) <- c("origin", "dest", "n", "era")
  fit_common <- MASS::loglm(
    n ~ era + origin + dest + origin:dest,
    data = df_counts)
  fit_diff <- MASS::loglm(
    n ~ era * origin * dest,
    data = df_counts)
  G2 <- fit_common$deviance - fit_diff$deviance
  df_diff <- fit_common$df - fit_diff$df
  p_val <- 1 - pchisq(G2, df_diff)
  list(
    counts = df_counts,
```

```

    G2 = G2,
    df = df_diff,
    p_value = p_val,
    fit_common = fit_common,
    fit_diff = fit_diff)
}

res <- compare_eras_markov(
  trans1_df = trans1,          # Era 1 transitions
  trans2_df = trans2,          # Era 2 transitions
  era1_label = "1860-1880",
  era2_label = "1881-1916",
  common_levels = c("upper_class", "middle_class", "lower_class_1"))

# Interpretation Guide
interpret_matrix_equality <- function(p, alpha = 0.05) {
  if (p < alpha) {
    paste0("Reject equality of transition matrices (p < $ ", alpha,
           "). Evidence of structural change between eras.")
  } else {
    paste0("Fail to reject equality of transition matrices (p >= $ ", alpha,
           "). Transition dynamics are statistically indistinguishable across eras.")
  }
}

# Run test and save
res <- compare_eras_markov(
  trans1_df = trans1,
  trans2_df = trans2,
  era1_label = "1860-1880",
  era2_label = "1881-1916",
  common_levels = c("upper_class", "middle_class", "lower_class_1")
)

```

11.1 Results

Test Statistic

Likelihood-ratio test (log-linear): $G^2(8) = 1.869$, $p = 0.9848$.

Interpretation

Fail to reject equality of transition matrices ($p \geq 0.05$). Transition dynamics are statistically indistinguishable across eras.

12 Appendix: Methodological Notes

```

# Balanced-panel of snapshot transitions

# Purpose:
#   Given a set of snapshot years {t_1, ..., t_T} and step length k,
#   construct transitions (X_t -> X_{t+k}) for a *balanced panel* of

```

```

# countries (stateabb).
#
# In Markov-chain terms, we observe a finite-state stochastic process
#  $X_t(u)$  taking values in a fixed state space  $S = \{upper\_class, semi, lower\_class\_1*\}$ 
# at discrete times  $t \in \{t_1, \dots, t_T\}$ . A balanced panel holds  $S$  (the set
# of units  $u$ ) fixed across snapshots so that transitions are between
# states rather than driven by changing sample composition
# (see Dezzani 2002 on state-level Markov models for the world-system).
#
# Inputs:
# data          : data frame with columns stateabb, year, class.
# snapshot_years: vector of years (e.g., 1860, 1865, 1704, 1875, 1880).
# step          : step size (e.g., 5 years).
# full_levels   : complete factor levels for class in this era,
#                 enforcing a fixed state space  $S$  across models.
#
# Steps (data-analytic):
# 1. Filter to snapshot years  $t_s \in \{t_1, \dots, t_T\}$ .
# 2. Within these years, keep only countries that appear in all
#    snapshot years  $\rightarrow$  ensures a balanced panel:
#     $U^* = \{u : u \text{ observed at every } t_s\}$ .
#    This isolates mobility within a fixed set of units, which is
#    critical for interpretable Markov mobility (Dezzani 2002; Babones
#    & Dezzani 2008).
# 3. For each  $u \in U^*$ , sort by year and compute lagged pairs
#     $X_t(u)$  and  $X_{t+k}(u)$ .
#    These give observed transitions between states  $i \rightarrow j$ .
# 4. Keep only pairs where  $(year - year\_prev) == step$ , so that we
#    construct k-step chains (Anderson & Goodman 1957).
# 5. Define  $step\_id = "t + t+k"$  to index the snapshot step  $s$  in
#    the 3-way table  $N_{\{sij\}}$ .
#
# Math:
# For each country  $u$  and snapshot  $s$  ( $t_s, t_{s+1} = t_s + k$ ):
#  $X_t(u)$           = class at time  $t_s$ 
#  $X_{t+k}(u)$       = class at time  $t_{s+1}$ 
#
# These transitions produce 3D counts  $N_{\{sij\}}$  used later in:
# - homogeneity tests across steps ( $N_{\{sij\}}$  as a 3-way table),
# - pooled transition matrix estimation  $N_{\{ij\}} = \sum_s N_{\{sij\}}$ .
#
# References:
# Dezzani (2002) "Measuring Transition and Mobility in the
# Hierarchical World-Economy", Journal of Regional Science.
# Babones & Dezzani (2008) "Mobility in the Modern World-Economy".
# Anderson & Goodman (1957) "Statistical Inference about Markov Chains".

# Balanced-panel attrition diagnostic

# Purpose:
# Compare class distributions for:

```

```

# - all countries in a year vs
# - countries retained in the balanced panel for the era.
#
# This checks whether panel restrictions introduce compositional
# bias in the Markov estimates (Dezzani 2002; Babones & Dezzani
# 2008).
#
# INTERPRETATION:
# A chi-square comparison of class distributions for all
# countries versus balanced-panel countries in the first
# snapshot year provides a diagnostic for potential attrition
# bias (Dezzani 2002; Babones & Dezzani 2008).

# 5-year pooled transition matrices

# Purpose:
# Pool transitions across steps to estimate:
# - transition probabilities  $\hat{P} = (P_{ij})$ ,
# - stationary distribution  $\hat{\pi}$ .
#
# Data:
# From all transitions (across steps), we construct a 2D table:
#  $N_{ij} = \sum_s N_{sij}$ ,
# where  $i$  = origin,  $j$  = destination.
#
# Estimation:
# Define:
#  $N_{ij}$  = total transitions from  $i$  to  $j$ ,
#  $N_{i+} = \sum_j N_{ij}$  (row totals).
# Then MLE for one-step Markov transition probabilities is:
#  $\hat{P}_{ij} = N_{ij} / N_{i+}$ . (Anderson & Goodman 1957)
#
# Stationary distribution:
# The stationary distribution satisfies:
#  $\hat{P} \hat{\pi} = \hat{\pi}$ ,  $\sum_i \hat{\pi}_i = 1$ .
#
# Numerically, we obtain  $\hat{\pi}$  as the dominant left eigenvector of  $\hat{P}$ :
# Let  $v$  be a right eigenvector of  $\hat{P}^T$ :
#  $\hat{P}^T v = \lambda v$ .
# If  $\lambda = 1$  is the largest eigenvalue (for an irreducible, aperiodic chain),
# then  $v$  (normalized to sum to 1) is the stationary distribution:
#  $\hat{\pi} = v / \sum_i v_i$ .
#
# References:
# Anderson & Goodman (1957) on ML estimation and asymptotic theory
# for discrete-state Markov chains.
# Bickelbach & Bode (2001) for applications of such matrices to
# regional/income convergence.

# 5-year step homogeneity tests

```

```

# Purpose:
#   Test whether transition probabilities  $P_{ij}$  are identical across
#   snapshot steps  $s$  (time-homogeneous Markov chain within an era).
#
#   Data structure:
#     - We construct a 3D contingency table  $N_{\{sij\}}$ :
#        $S$  = step (snapshot interval),
#        $O$  = origin class,
#        $D$  = destination class.
#     - In log-linear form, this is a standard three-way table  $S \times O \times D$ 
#       (Bishop, Fienberg & Holland 1975/2007; Goodman 1971).
#
#   Model (log-linear representation):
#     Let  $\_ \{sij\} = E[N_{\{sij\}}]$  denote expected counts under a given model.
#
#      $H_0$ : Homogeneous Markov model (OD association invariant in  $s$ )
#        $\log \_ \{sij\} = \_ s + \_ i + \_ j + \_ \{ij\} \{ (OD) \}$ 
#       → includes:
#         - main effects for step, origin, destination,
#         - origin-destination association  $\_ \{ij\} \{ (OD) \}$  *not* varying
#           with  $s$ , corresponding to a time-homogeneous chain.
#
#      $H_1$ : Fully heterogeneous model
#        $\log \_ \{sij\} =$  + all interactions among  $(S, O, D)$ 
#                       = +  $\_ s + \_ i + \_ j$ 
#                         +  $\_ \{si\} + \_ \{sj\} + \_ \{ij\} + \_ \{sij\}$ .
#       → OD association allowed to differ by step via  $\_ \{sij\}$ .
#
#   Test statistic:
#      $G^2 = \text{deviance}(H_0) - \text{deviance}(H_1)$ ,
#      $df = df(H_0) - df(H_1)$ ,
#     p-value from  $\chi^2_{df}$ .
#
#   Here, deviance is the standard log-likelihood ratio statistic:
#      $G^2(\text{model}) = 2 \sum \_ \{sij\} n_{\{sij\}} \log( n_{\{sij\}} / \_ \{sij\} )$ ,
#     which is  $2 \times (\text{sample size})$  times a Kullback-Leibler divergence between
#     observed and fitted cell probabilities (Kullback & Leibler 1951;
#     Kullback, Kupperman & Ku 1962). Under regularity conditions,  $G^2$ 
#     is asymptotically  $\chi^2$  (Anderson & Goodman 1957; Bishop et al. 1975).
#
#   Relation to Markov-chain diagnostics:
#     - This is essentially the log-linear version of testing whether
#       transition matrices differ across time slices, analogous to the
#       homogeneity tests in Bickelbach & Bode's (2001) discussion of
#       Markov chains for income distributions.

# Long-interval homogeneity tests

# Purpose:
#   Build a *long-interval* transition matrix for each era and test

```

```

# whether it is consistent with a stationary/quasi-symmetric structure.
#
# Here we treat the initial and final snapshots (start_year, end_year)
# as two "waves" in a square contingency table:
#   rows = start-year classes,
#   cols = end-year classes.
#
# Data construction:
#   For each country u:
#     class_start(u) = class in start_year,
#     class_end(u)   = class in end_year.
#
# We then build the 2-way table:
#   L_{ij} = # of countries with class_start = i and class_end = j.
#
# Log-linear testing:
#   We compare:
#   - H0: structured (origin + dest) model:
#     log _{ij} = _i + _j,
#     which implies that rows and columns are independent up to
#     row and column effects (a kind of "quasi-stationary" structure),
#     closely related to mobility models and quasi-symmetry ideas.
#
#   - H1: saturated model:
#     log _{ij} = _{ij},      (free mean for each cell).
#
# LR test:
#   G^2_long = dev(H0) - dev(H1),
#   df_long  = df(H0) - df(H1),
#   p-value from ^2_{df_long}.
#
# Information-theoretic view:
#   As in the homogeneity test, the deviance is 2*(sample size) times
#   a Kullback-Leibler divergence between empirical L_{ij}/N and
#   the fitted ^_{ij}/N (Kullback & Leibler 1951; Kullback, Kupperman
#   & Ku 1962; Bishop et al. 1975).
#
# References:
#   Kullback, Kupperman & Ku (1962) "Tests for Contingency Tables and
#   Markov Chains", Technometrics.
#   Bishop, Fienberg & Holland (1975/2007) "Discrete Multivariate Analysis".
#   Goodman (1971, Technometrics) on log-linear models for multidimensional
#   tables.

# Markov pairs X_t, X_{t+1} (0 + 1 order)

# Purpose:
#   Construct ordered pairs (X_t, X_{t+1}) for estimating and testing
#   *zero-order vs first-order* dependence in a discrete-state process.
#
#   Specifically, the pair table supports the hypothesis test:

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#
#   H0 (0th order / independence):    $X_{t+1} \perp X_t$ 
#   H1 (1st order / Markov):          $X_{t+1}$  depends on  $X_t$ 
#
#   In a Markov-chain setting, H1 corresponds to the standard
#   first-order dependence assumption used to estimate a transition
#   matrix  $P$  with entries  $P_{ij} = \Pr(X_{t+1}=j \mid X_t=i)$ , as developed in
#   Anderson & Goodman (1957, Annals of Mathematical Statistics).
#
#   In contingency-table terms, we represent transitions as a 2-way table
#    $N_{ij}$  of origin ( $i = X_t$ ) by destination ( $j = X_{t+1}$ ) counts and fit
#   hierarchical log-linear models:
#
#   - Independence (0th order):
#        $\log \pi_{ij} = \mu + \mu_i^{(0)} + \mu_j^{(D)}$ 
#       (no origin-destination association)
#
#   - Association (1st order):
#        $\log \pi_{ij} = \mu + \mu_i^{(0)} + \mu_j^{(D)} + \mu_{ij}^{(OD)}$ 
#
#   The likelihood-ratio deviance  $G^2$  compares these models:
#
#        $G^2 = 2 \sum_{ij} n_{ij} \log(n_{ij} / \hat{\pi}_{ij})$ 
#
#   which is  $2N$  times a Kullback-Leibler divergence between empirical
#   and fitted cell probabilities (Kullback & Leibler 1951; Kullback,
#   Kupperman & Ku 1962). Under standard regularity conditions,  $G^2$ 
#   is asymptotically  $\chi^2$  (Anderson & Goodman 1957; Bishop, Fienberg &
#   Holland 1975/2007; Goodman 1971).
#
# Inputs:
# data           : data frame with columns stateabb, year, class.
# snapshot_years: ordered vector of observation years.
# step           : spacing constraint between consecutive snapshots
#                 (e.g., 7; 8; or a vector like c(7,7,6)).
# full_levels    : factor levels for class (fixed state space  $S$ ).
#
# Output:
# data frame with columns:
#   stateabb, year, class_prev, class_curr
# restricted to balanced-panel countries and exact step spacing.

# Markov triples  $X_{t-1}, X_t, X_{t+1}$  (1 + 2 order)

# Purpose:
# Construct triples  $(X_{t-1}, X_t, X_{t+1})$  for testing the
# first-order Markov property:
#
#   H0:  $X_{t+1} \perp X_{t-1} \mid X_t$ 
#
# Following Anderson & Goodman (1957, Annals of Mathematical

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# Statistics) and the log-linear representation of Markov chains
# in Bishop, Fienberg & Holland (1975/2007, Discrete Multivariate
# Analysis) and Goodman (1971, Technometrics).
#
# Inputs:
#   data           : df with stateabb, year, class
#   snapshot_years : ordered vector of observation years
#   step           : spacing between snapshots (e.g., 7, 8)
#   full_levels    : factor levels for class
#
# Output:
#   data frame with columns:
#     stateabb, year, class_lag2, class_lag1, class_curr
#   restricted to balanced-panel countries and exact step spacing.
#
# References:
#   Anderson, T.W. & Goodman, L.A. (1957). "Statistical Inference about
#     Markov Chains." Annals of Mathematical Statistics.
#   Bishop, Y.M.M., Fienberg, S.E., & Holland, P.W. (1975/2007).
#     Discrete Multivariate Analysis. MIT Press.
#   Goodman, L.A. (1971). "The Analysis of Multidimensional Contingency
#     Tables." Technometrics.
#   Kullback, S. & Leibler, R.A. (1951). "On Information and Sufficiency."
#     Annals of Mathematical Statistics.
#   Kullback, S., Kupperman, M., & Ku, H. (1962). "Tests for Contingency
#     Tables and Markov Chains." Technometrics.

# 0 vs 1 order tests ( $X_{t-1}$ ,  $X_t$ )

# Purpose:
#   Test the first-order Markov property:
#
#    $H_0: X_{t+1} \sim X_{t-1} \mid X_t$ 
#
#   using a three-way contingency table:
#    $L = X_{t-1}$ ,  $M = X_t$ ,  $N = X_{t+1}$ .
#
# Log-linear models:
#
#   Let  $n_{ijk} = E[N_{ijk}]$  = expected counts for  $L=i$ ,  $M=j$ ,  $N=k$ .
#
#    $H_0$  (first-order Markov):
#    $\log n_{ijk} =$ 
#        $+ \alpha_i^L + \alpha_j^M + \alpha_k^N$ 
#        $+ \alpha_{ij}^{LM} + \alpha_{jk}^{MN}$ 
#
#   → includes L-M and M-N interactions, but no L-N or L-M-N
#     interactions, corresponding to  $X_{t+1} \sim X_{t-1} \mid X_t$ .
#
#    $H_1$  (saturated three-way):
#    $\log n_{ijk} =$  + all main effects + all interactions

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#           = L * M * N in loglm() formula notation.
#
# LR test:
#   G^2 = dev(H0) - dev(H1),
#   df  = df(H0) - df(H1),
#   p   = 1 - F_2(G^2; df).
#
# References:
#   Anderson & Goodman (1957) "Statistical Inference about Markov
#   Chains", Annals of Mathematical Statistics.
#   Bishop, Fienberg & Holland (1975/2007) Discrete Multivariate
#   Analysis, MIT Press.
#   Goodman (1971) "The Analysis of Multidimensional Contingency
#   Tables", Technometrics.

# 1 vs 2 order tests (X_{t-2}, X_{t-1}, X_t)

# Purpose:
#   Test whether X_{t} depends on X_{t-2k} given X_{t-k}:
#
#       H0 (1st-order): N ~ L | M
#       H1 (2nd-order): N depends on both M and L
#
# Models (log-linear):
#
#   H0: log _LMN = L + M + N + L:M + M:N
#   H1: log _LMN = L * M * N
#
# LR statistic:
#   G^2 = dev(H0) - dev(H1)
#   df  = df(H0) - df(H1)
#
# Citations:
#   Anderson & Goodman (1957)
#   Bishop, Fienberg & Holland (1975/2007)
#   Goodman (1971)

# Ergodicity diagnostic (irreducibility & aperiodicity)

# Purpose:
#   Provide diagnostics for:
#   - irreducibility (all states communicate),
#   - aperiodicity (no deterministic cycles),
#   using powers of the transition matrix P.
#
# Approach:
#   - Construct adjacency matrix A with A_ij = 1 if P_ij > 0.
#   - Compute reachability via powers of A.
#   - For aperiodicity, check whether some power of P has strictly

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#     positive diagonal entries for all states.
#
# Interpretation:
#   If some  $k$  max_power has all entries of  $A^k > 0$ , the chain
#   is regular (thus irreducible and aperiodic).
#
# References:
#   Anderson & Goodman (1957) on long-run properties of Markov chains.
#   Standard Markov chain theory for ergodicity criteria.

# Equality of transition matrices between eras

# Purpose:
#   Compare transition structures between Era 1 and Era 2 on the
#   shared class set (upper_class, middle_class, lower_class_1), using a
#   4-way contingency table with factors:
#
#     Era (E), Origin (O), Destination (D)
#
# Log-linear setup:
#   H0: common Markov structure across eras:
#        $\log \pi_{eij} = \mu + \alpha_e^E + \alpha_i^O + \alpha_j^D + \alpha_{ij}^{OD}$ 
#
#   H1: era-specific OD association:
#        $\log \pi_{eij} = \mu + \text{all main effects} + \text{all interactions}$ 
#            $= E * O * D$ 
#
# LR test:
#    $G^2 = dev(H0) - dev(H1)$ ,
#    $df = df(H0) - df(H1)$ .
#
# References:
#   Bishop, Fienberg & Holland (1975/2007), Goodman (1971),
#   Bickelbach & Bode (2001) on comparing Markov structures
#   across groups/periods.

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